

5/11

① rigid body kinematics

② rigid body kinetics : 힘이 가해졌을 때 어떻게 운동 할까?

③ control : 특정 자세로 어떻게 돌아갈 것인가?

* Kinematics

1. point mass

2. rigid body $\rightarrow$ cosine matrix, rotation matrix, $\oplus$ Classic 한 yaw, pitch, roll rotation Vector. quaternions?

no singularity

* Particle kinematics

$$r = x\hat{e}_1 + y\hat{e}_2 + z\hat{e}_3$$

$$= r\hat{e}_r$$

$$= \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$\rightarrow$ Vector를 matrix 형태로 표현한 것

$\star$ 모든 Vector는 matrix로 나타낼 수 있지만, 모든 matrix는 vector로 나타낼 수 없다.

$\star$ 이 자체는 좌표계가 정해지지 않은 vector로 계산 가능

특히 Vector의 의미에 대해 생각해야 한다. $\vec{q} = \vec{r} + \vec{p}$ 라는 식에서 r, p, q는 같은 frame 안에 있지 않을 수도 있다.

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} + \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}$$

여기면 frame이 다르므로 계산은 되지만,

meaningful 하지는 않음. 그럼에도 매트랩은 이런 계산을 하는 문제. 따라서 이것이 의미는

가지려면 common frame이 있도록 해야함.

1
모두

coordinate frame: $B: \{b_1, b_2, b_3\}$

shorthand notation으로 쓴다.

ex)

$$\begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} : \text{matrix}$$

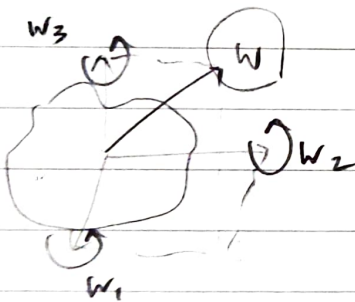
$$E \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} \equiv (5m)\hat{e}_1 + (3m)\hat{e}_2 + (2m)\hat{e}_3$$

matlab에서 vector끼리 더할 때는 같은 frame이 있도록 해야함. 만약 B, E 가 있는 것들은 E 에 rotation matrix로 같은 frame으로 바꾼 후 더함.

Vector $\rightarrow$ matrix (0)

matrix $\rightarrow$ vector (X)

* Angular velocity vector.



$$\vec{w} = w_1 \hat{b}_1 + w_2 \hat{b}_2 + w_3 \hat{b}_3$$

이 rigid body의 $\vec{w}_{net}$ 이 외부 inertial frame N 에 대해 상대적으로 어떤 상황인지 알아내는 것이 중요!

이 역할을 자이로스코프가 함.

Angular velocity relative to inertial frame

$$w_{B/O} = w_{B/N} - w_{O/N}$$

(B relative to orbit) (B relative to N) (O relative to N)

Positive rotation: right hand rule은 따름.

* vector differentiation

what is a derivative of vector? : 어느 좌표계에서 바라보느냐에 따라 0 or non zero.

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r}$$

$$\mathbf{r} = r_1 \hat{\mathbf{b}}_1 + r_2 \hat{\mathbf{b}}_2 + r_3 \hat{\mathbf{b}}_3$$

$$\mathbf{W}_{B/N} = w_1 \hat{\mathbf{b}}_1 + w_2 \hat{\mathbf{b}}_2 + w_3 \hat{\mathbf{b}}_3$$

$$\frac{B}{d} \frac{d}{dt}(\mathbf{r}) = \dot{r}_1 \hat{\mathbf{b}}_1 + \dot{r}_2 \hat{\mathbf{b}}_2 + \dot{r}_3 \hat{\mathbf{b}}_3 \rightarrow \text{B frame에서 볼 때는 easily calculated.}$$

$$\frac{N}{d} \frac{d}{dt}(\mathbf{r}) = \dot{r}_1 \hat{\mathbf{b}}_1 + \dot{r}_2 \hat{\mathbf{b}}_2 + \dot{r}_3 \hat{\mathbf{b}}_3 + r_1 \frac{d\hat{\mathbf{b}}_1}{dt} + r_2 \frac{d\hat{\mathbf{b}}_2}{dt} + r_3 \frac{d\hat{\mathbf{b}}_3}{dt}$$

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r} \text{ 이므로 } \frac{N}{d} \frac{d}{dt}(\hat{\mathbf{b}}_i) = \mathbf{W}_{B/N} \times \hat{\mathbf{b}}_i$$

$$r_1 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_1) + r_2 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_2) + r_3 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_3)$$

$$= \mathbf{W}_{B/N} \times (r_1 \hat{\mathbf{b}}_1 + r_2 \hat{\mathbf{b}}_2 + r_3 \hat{\mathbf{b}}_3)$$

$$= \mathbf{W}_{B/N} \times \mathbf{r}$$

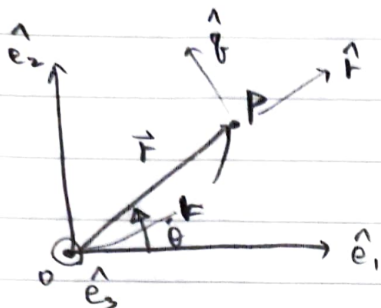
$$\therefore \frac{N}{d} \frac{d}{dt}(\mathbf{r}) = \frac{B}{d} \frac{d}{dt}(\mathbf{r}) + \mathbf{W}_{B/N} \times \mathbf{r}$$

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r}$$

5/12

Kinematics: 운동학

Kinetics: 역학



< P를 나타내는 방법 >

- ① write vector
- ② define frame
- ③ get W's
- ④ differentiate

$P: \{ \hat{f}, \hat{g}, \hat{e}_3 \} \rightarrow$ coordinate frame.

$W_{P/E} = \dot{\theta} \hat{e}_3$: P가 Planar로 회전한다고 가정할 시.

inertial frame : non-accelerating frame

$\dot{\vec{r}} \equiv \frac{d}{dt}(\vec{r})$ 그런데 frame을 N에서 P로 바꾸는데 사실 더 편함

$$\frac{d}{dt}(\vec{r}) = \frac{d}{dt}(\vec{r} \hat{f}) = \dot{\vec{r}} \frac{d}{dt}(\hat{f}) + \vec{r} \frac{d}{dt}(\hat{f}) = \dot{\vec{r}} \hat{f}$$

$$\dot{\vec{r}} = \frac{d}{dt}(\vec{r}) = \underbrace{(\dot{\vec{r}} \hat{f})}_{P \text{ frame}} + W_{P/E} \times \vec{r}$$

$$= \dot{\vec{r}} \hat{f} + \dot{\theta} \hat{e}_3 \times \vec{r}$$

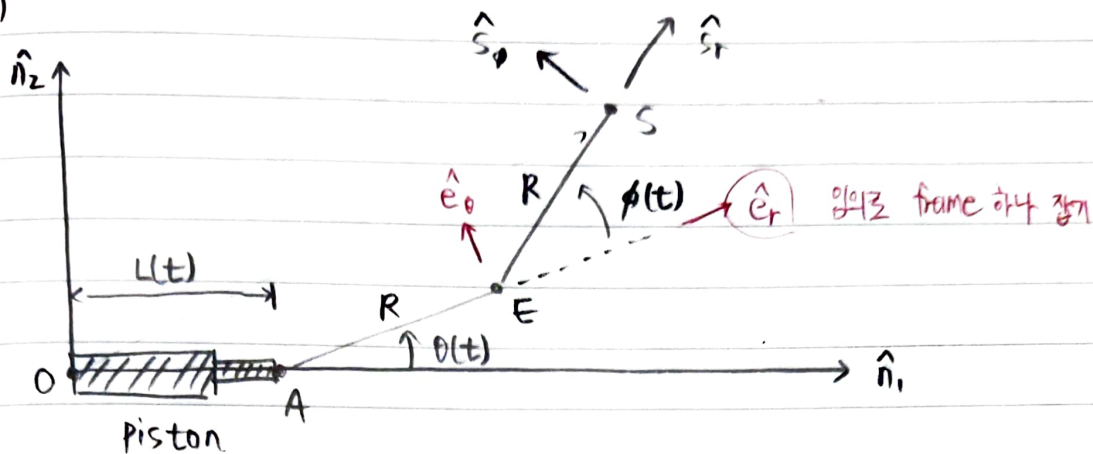
$$= \dot{\vec{r}} \hat{f} + \dot{\theta} \hat{g} \quad \dots \quad \text{Use rotating frame}$$

GNC에서 많이 사용.

만약 $\vec{r} = a\hat{a}_1 + b\hat{b}_1$ 라 한다면 가장 lazy하고 easy한 방법으로 풀이하는 방법은

$$\dot{\vec{r}} = \left\{ \frac{d}{dt}(a\hat{a}_1) + W_{A/N} \times a\hat{a}_1 \right\} + \left\{ \frac{d}{dt}(b\hat{b}_1) + W_{B/N} \times b\hat{b}_1 \right\}$$

ex)



(inertial)
① S의 Velocity를 구하라.

Position of A: $L(t) \hat{n}_1$

Position of E: $L(t) \hat{n}_1 +$

$$W_{E/N} = \dot{\theta}$$

$$W_{S/N} = \dot{\theta} + \dot{\phi}$$

각속도는 벡터임!!!

$$\frac{d}{dt}(\vec{s}) = \frac{d}{dt}(\vec{s}) + W_{E/N} \times \vec{s}$$

$$\dot{\vec{s}} = \dot{\phi} + \dot{\theta} \times \vec{s}$$

$$\vec{r} = L \hat{n}_1 + R \hat{e}_r + R \hat{s}_r = \vec{r}_{S/N}$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$W_{E/N} = \dot{\theta} \hat{n}_3$$

$$E: \{ \hat{e}_r, \hat{e}_\theta, \hat{n}_3 \}$$

$$W_{S/E} = \dot{\phi} \hat{n}_3$$

$$S: \{ \hat{s}_r, \hat{s}_\theta, \hat{n}_3 \}$$

$$W_{S/N} = W_{E/N} + W_{S/E} \\ = (\dot{\theta} + \dot{\phi}) \hat{n}_3$$

$$\dot{\vec{r}} = \frac{d}{dt}(\vec{r}) = \frac{d}{dt}(L \hat{n}_1) + \left\{ \frac{d}{dt}(R \hat{e}_r) + W_{E/N} \times (R \hat{e}_r) \right\} + \left\{ \frac{d}{dt}(R \hat{s}_r) + W_{S/N} \times (R \hat{s}_r) \right\}$$

$$= \dot{L} \hat{n}_1 + \{ \dot{\theta} \hat{n}_3 \times (R \hat{e}_r) \} + \{ (\dot{\theta} + \dot{\phi}) \hat{n}_3 \times (R \hat{s}_r) \}$$

$$= \dot{L} \hat{n}_1 + R \dot{\theta} \hat{e}_\theta + R(\dot{\theta} + \dot{\phi}) \hat{s}_\theta$$

② S의 inertial acceleration을 구하라

$$\dot{\mathbf{r}} = \dot{L} \hat{\mathbf{n}}_1 + R \dot{\theta} \hat{\mathbf{e}}_\theta + R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi$$

$$\ddot{\mathbf{r}} = \frac{d}{dt}(\dot{\mathbf{r}})$$

$$= \frac{d}{dt}(\dot{L} \hat{\mathbf{n}}_1) + \frac{d}{dt}(R \dot{\theta} \hat{\mathbf{e}}_\theta) + \frac{d}{dt}(R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi)$$

$$= \ddot{L} \hat{\mathbf{n}}_1 + \left[\begin{array}{c} + R \ddot{\theta} \hat{\mathbf{e}}_\theta \\ + W_{E/N} \times (R \dot{\theta} \hat{\mathbf{e}}_\theta) \end{array} \right] + \left[\begin{array}{c} + R(\ddot{\theta} + \ddot{\phi}) \hat{\mathbf{s}}_\phi \\ + W_{S/N} \times (R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi) \end{array} \right]$$

$$= \ddot{L} \hat{\mathbf{n}}_1 + R \ddot{\theta} \hat{\mathbf{e}}_\theta - R \dot{\theta}^2 \hat{\mathbf{e}}_r + R(\ddot{\theta} + \ddot{\phi}) \hat{\mathbf{s}}_\phi - R(\dot{\theta} + \dot{\phi})^2 \hat{\mathbf{s}}_r$$

↳ sin, cos로 나누어 하는 것보다 훨씬 간편, 유용!

③ S위치에서 볼 A위치의 움직임

$$\mathbf{r}_{A/S} = -R \hat{\mathbf{e}}_r - R \hat{\mathbf{s}}_r$$

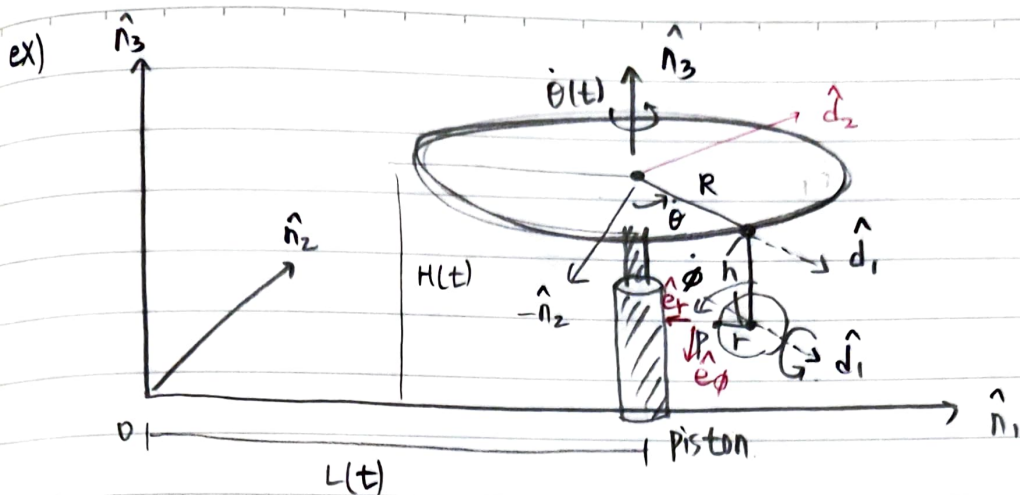
very easy

$$\frac{d}{dt}(\mathbf{r}_{A/S}) = -W_{E/S} \times R \hat{\mathbf{e}}_r$$

계산자 하는 것

$$= \dot{\phi} \hat{\mathbf{n}}_3 \times R \hat{\mathbf{e}}_r$$

$$= R \dot{\phi} \hat{\mathbf{e}}_\theta$$



Process 2

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$D: \{ \hat{d}_1, \hat{d}_2, \hat{n}_3 \}$$

$$E: \{ \hat{e}_r, \hat{e}_\phi, \hat{d}_1 \}$$

Velocity를 찾는거까 구한 것은 $r_{P/O}$ 정도 된다.

$$r_{P/O} = L \hat{n}_1 + H \hat{n}_3 + R \hat{d}_1 - h \hat{n}_3 + r \hat{e}_r$$

Process 1

$$W_{D/N} = \dot{\theta} \hat{n}_3$$

$$W_{E/D} = \dot{\phi} \hat{d}_1$$

$$W_{E/N} = \dot{\phi} \hat{d}_1 + \dot{\theta} \hat{n}_3$$

Process 3

Process 4: differentiate.

$$\dot{r}_{P/O} = \dot{L} \hat{n}_1 + \dot{H} \hat{n}_3 + \{ W_{D/N} \times R \hat{d}_1 \} + \{ W_{E/N} \times r \hat{e}_r \}$$

$$= \dot{L} \hat{n}_1 + \dot{H} \hat{n}_3 + R \dot{\theta} \hat{d}_2 + r \dot{\phi} \hat{e}_\phi + \underbrace{(\dot{\theta} \hat{n}_3 \times r \hat{e}_r)}$$

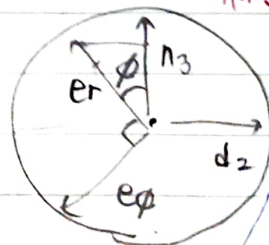
$$\hat{n}_3 \perp \hat{e}_r \text{ 이 아님}$$

$$\rightarrow \hat{n}_3 \times \hat{e}_r$$

$$\rightarrow (\cos \phi \hat{n}_2 - \sin \phi \hat{d}_2)$$

$$= \sin \phi \hat{d}_1$$

이 라장 계산을
모르겠다.

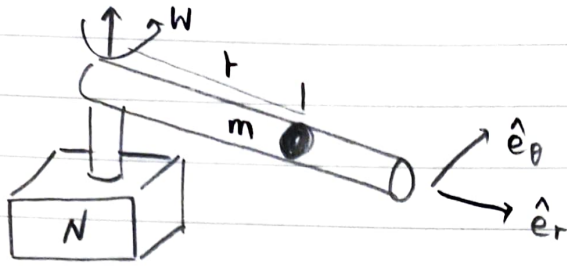


$$= \dot{L} \hat{n}_1 + \dot{H} \hat{n}_3 + R \dot{\theta} \hat{d}_2 + r \dot{\phi} \hat{e}_\phi + r \dot{\theta} \sin \phi \hat{d}_1$$

이 경우에는 행렬 연산을
해야한다.

<concept check 3> 5/13

1)



$$\vec{r} = r\hat{e}_r \quad \text{--- ①}$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

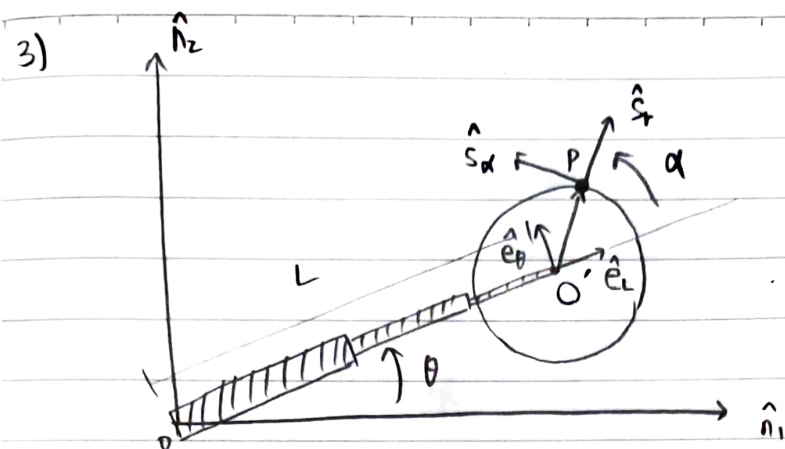
$$E: \{ \hat{e}_r, \hat{e}_\theta, \hat{n}_3 \} \quad \text{--- ②}$$

$$W_{E/N} = W\hat{n}_3, \quad W = \dot{\theta}$$

$$\begin{aligned} \frac{d}{dt}(\vec{r}) &= \dot{r}\hat{e}_r + W_{E/N} \times r\hat{e}_r \\ &= \dot{r}\hat{e}_r + rW\hat{e}_\theta = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta = \dot{\vec{r}} \end{aligned}$$

2) Acceleration

$$\begin{aligned} \frac{d}{dt}(\dot{\vec{r}}) &= \left\{ \frac{d}{dt}(\dot{r}\hat{e}_r) + W_{E/N} \times \dot{r}\hat{e}_r \right\} + \left\{ \frac{d}{dt}(r\dot{\theta}\hat{e}_\theta) + W_{E/N} \times r\dot{\theta}\hat{e}_\theta \right\} \\ &= \left\{ \ddot{r}\hat{e}_r + \dot{r}W\hat{e}_\theta \right\} + \left\{ r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\hat{e}_\theta - rW\dot{\theta}\hat{e}_r \right\} \\ &= (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{e}_\theta \end{aligned}$$



$$\vec{P} = L\hat{e}_L + r\hat{s}_r \quad \dots \quad (1)$$

$$W_{E/N} = \dot{\theta} \hat{n}_3$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$W_{S/E} = \dot{\alpha} \hat{n}_3$$

$$E: \{ \hat{e}_L, \hat{e}_\theta, \hat{n}_3 \}$$

$$W_{S/N} = (\dot{\theta} + \dot{\alpha}) \hat{n}_3 \quad \dots \quad (2)$$

$$S: \{ \hat{s}_r, \hat{s}_\alpha, \hat{n}_3 \} \quad \dots \quad (2)$$

$$\begin{aligned} \frac{N_d}{dt}(\vec{P}) &= \{ L\dot{\hat{e}}_L + W_{E/N} \times L\hat{e}_L \} + \{ W_{S/N} \times r\hat{s}_r \} \\ &= L\dot{\theta} \hat{e}_\theta + L\dot{\theta} \hat{e}_\theta + r(\dot{\theta} + \dot{\alpha}) \hat{s}_\alpha = \dot{\vec{P}} \end{aligned}$$

4) Acceleration

$$\begin{aligned} \frac{N_d}{dt}(\dot{\vec{P}}) &= \{ W_{E/N} \times L\dot{\hat{e}}_L \} + \{ (L\ddot{\theta} + L\dot{\theta}) \hat{e}_\theta + W_{E/N} \times L\dot{\theta} \hat{e}_\theta \} \\ &\quad + \{ r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha + W_{S/N} \times r(\dot{\theta} + \dot{\alpha}) \hat{s}_\alpha \} \\ &= L\ddot{\theta} \hat{e}_\theta + (L\ddot{\theta} + L\dot{\theta}) \hat{e}_\theta - L\dot{\theta}^2 \hat{e}_L + r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha - r(\dot{\theta} + \dot{\alpha})^2 \hat{s}_r \\ &= (2L\ddot{\theta} + \cancel{L\dot{\theta}}) \hat{e}_\theta - L\dot{\theta}^2 \hat{e}_L + \cancel{r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha} - r(\dot{\theta} + \dot{\alpha})^2 \hat{s}_r \end{aligned}$$

constant rate

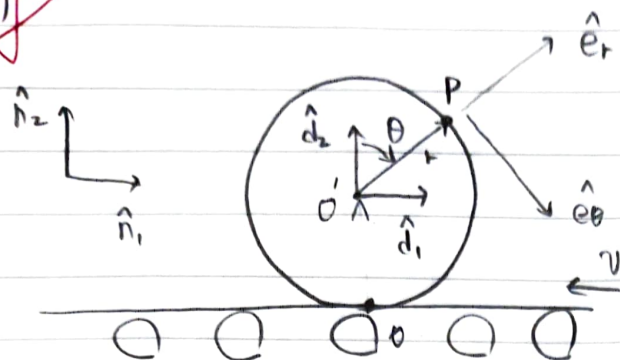
5) 아직 모르겠음

inertial acceleration이

어떻게 존재?



좌표계만 inertial, 그 안의 object는 가속도를 가질 수 있음



$$\vec{P} = r\hat{e}_r + vt\hat{d}_1 + r\hat{d}_2$$

$$W_{D/N} = 0$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$W_{E/N} = W_{E/D} = -\dot{\theta}\hat{n}_3$$

$$D: \{ \hat{d}_1, \hat{d}_2, \hat{n}_3 \}$$

$$E: \{ \hat{e}_r, \hat{e}_\theta, -\hat{n}_3 \} \quad \text{--- ②}$$

각도에 따른 회전속도가 세로 $\hat{e}_3$ 를 향하게

$$\begin{aligned} \frac{W_D}{dt}(\vec{P}) &= \{ W_{E/N} \times r\hat{e}_r \} + \{ v\hat{d}_1 + W_{E/N} \times vt\hat{d}_1 \} + \{ W_{D/N} \times r\hat{d}_2 \} \\ &= r\dot{\theta}\hat{e}_\theta + v\hat{d}_1 - vt\dot{\theta}\hat{d}_2 \end{aligned}$$

6)

$$W_{D/N} \times (r\dot{\theta} + v) - W_{E/N} \times r\dot{\theta}\hat{e}_\theta$$

$$= \dot{\theta}\hat{n}_3 \times r\dot{\theta}\hat{e}_\theta = r\dot{\theta}^2\hat{e}_r$$

$$W_{D/N} \times (r\dot{\theta} + v)\hat{d}_1 + W_{E/N} \times 2r\dot{\theta}\hat{e}_\theta$$

=

$$E: \{ \hat{e}_r, \hat{e}_\theta, \hat{e}_\phi \}$$

$$S: \{ \hat{s}_r, \hat{s}_\theta, \hat{s}_z \}$$

$$W_{E/N} = \dot{\phi} \hat{n}_3$$

$$W_{S/E} = \dot{\theta} \hat{S}_3$$

$$W_{S/N} = \dot{\phi} \hat{n}_3 + \dot{\theta} \hat{s}_3$$

$$\vec{p} = R\hat{e}_r + t\hat{s}_r$$

$$\frac{N_d}{dt}(\vec{p}) = \{W_{E/N} \times R \hat{e}_r\} + \{W_{S/N} \times r \hat{s}_r\}$$

$$= R\dot{\phi}\hat{e}_\phi + r\dot{\phi}\sin\theta\hat{s}_3 + r\dot{\theta}\hat{s}_\theta$$

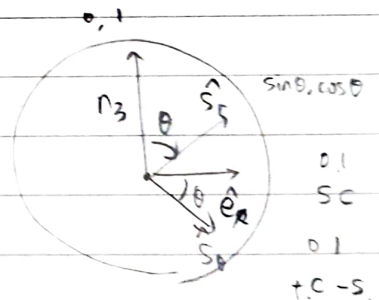
$$b) \{ R\ddot{\phi}\hat{e}_\phi + W_{E/N} \times R\dot{\phi}\hat{e}_\phi \} + \{ (r\ddot{\theta}\sin\theta + r\dot{\theta}\dot{\phi}\cos\theta)\hat{S}_3 + W_{S/N} \times r\dot{\theta}\sin\theta\hat{S}_3 \}$$

$$+ \{ r \ddot{\theta} \hat{S}_\theta + W_{S/N} \times r \dot{\theta} \hat{S}_\theta \}$$

$$= R\ddot{\phi}\hat{e}_\phi - R\dot{\phi}^2\hat{e}_r + (r\ddot{\phi}\sin\theta + r\dot{\theta}\dot{\phi}\cos\theta)\hat{s}_3$$

$$- \ddot{\theta}^2 \sin \theta \hat{e}_\phi + \ddot{\theta} \hat{s}_\theta + \dot{\theta} \dot{\phi} \cos \theta \hat{s}_3 - \dot{\theta}^2 \hat{s}_r$$

$$= (R\ddot{\phi} - r\dot{\phi}^2 \sin\theta) \hat{e}_\theta + \hat{e}_r (-r\dot{\theta}\dot{\phi} \cos\theta - R\ddot{\theta} - r\ddot{\phi} \sin\theta - r\dot{\theta}\dot{\phi} \cos\theta) + r\dot{\theta} \hat{S}_\theta - r\dot{\theta}^2 \hat{S}_r$$



$$\hat{n}_3 \times \hat{s}_t = \hat{s}_3 \times \sin \theta$$

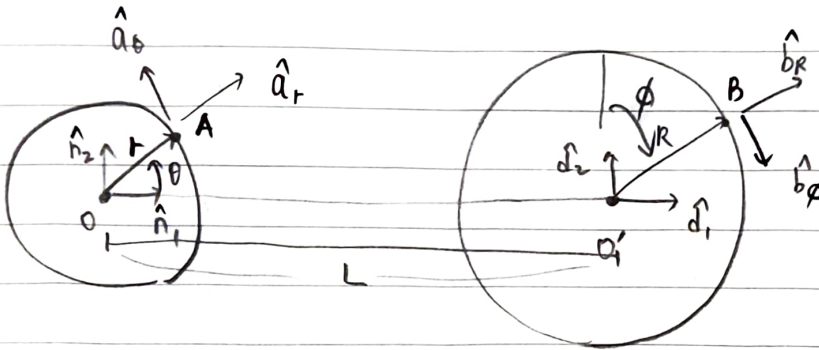
$$9) \quad \vec{r}_m = h \hat{n}_3 - V_0 t \hat{n}_2$$

$$\dot{\vec{r}}_m = -V_0 \hat{n}_2$$

$$\vec{V}_{P/m} = \dot{\vec{r}} = \vec{V}_P - \vec{V}_m$$

$$= R\dot{\phi} \hat{e}_\phi + r\dot{\phi} \sin\theta \hat{s}_3 + r\dot{\theta} \hat{s}_\theta + V_0 \hat{n}_2$$

10)



$$\vec{B} = R \hat{b}_R + L \hat{d}_1, \quad W_{B/N} = -\dot{\phi} \hat{n}_3$$

$$W_{B/N} \times R \hat{b}_R = R \dot{\phi} \hat{b}_\phi = \vec{v}_B$$

$$A: \{ \hat{a}_r, \hat{a}_\theta, \hat{n}_3 \}$$

$$O: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$D: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$B: \{ \hat{b}_R, \hat{b}_\phi, -\hat{n}_3 \}$$

$$11) \quad W_{B/N} \times R \dot{\phi} \hat{b}_\phi$$

$$= -R \dot{\phi}^2 \hat{b}_R$$

$$W_{B/A} = -\dot{\phi} \hat{n}_3 - \dot{\theta} \hat{n}_3$$

$$12) \quad \vec{A} = r \hat{a}_r, \quad W_{A/N} = \dot{\theta} \hat{n}_3$$

$$\vec{r}_{B/A} = R \hat{b}_R + L \hat{d}_1 - r \hat{a}_r$$

$$\begin{aligned} \frac{d}{dt} (\vec{r}_{B/A}) &= \{ W_{B/A} \times R \hat{b}_R \} + \{ W_{N/A} \times L \hat{d}_1 \} \\ &= R(\dot{\theta} + \dot{\phi}) \hat{b}_\phi - L \dot{\theta} \hat{d}_2 \end{aligned}$$

$$13) \quad W_{B/N} = -\dot{\phi} \hat{n}_3$$

$${}^N \frac{d}{dt} (-\dot{\phi} \hat{n}_3) = {}^B \frac{d}{dt} (-\dot{\phi} \hat{n}_3) + W_{B/N} \times (-\dot{\phi} \hat{n}_3)$$

★ ~~문제~~ 해결

15) B frame과 N frame에서 볼 때 항상 같지? $W_{B/N}$ or $W_{N/B}$

Because
$${}^N \frac{d}{dt} () = {}^B \frac{d}{dt} () + W_{B/N} \times ()$$

↗ 0

↓

Cross product 가 0이 되어
앞의 항들만 남기 때문

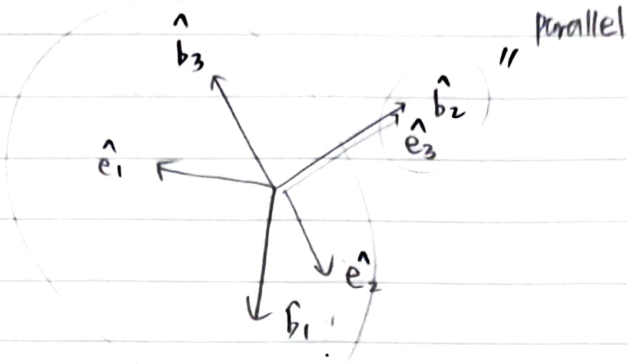
16)

$$r = r_1 \hat{b}_1 + r_2 \hat{b}_2 + r_3 \hat{b}_3 \equiv {}^B \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}$$

$$\underbrace{{}^B \frac{d}{dt} (r)} = \dot{r}_1 \hat{b}_1 + \dot{r}_2 \hat{b}_2 + \dot{r}_3 \hat{b}_3 = {}^B \begin{pmatrix} \dot{r}_1 \\ \dot{r}_2 \\ \dot{r}_3 \end{pmatrix} = \underbrace{\frac{d}{dt} ({}^B r)}$$

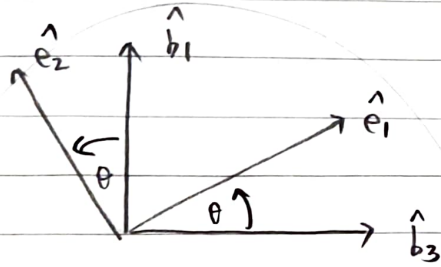
큰 의미 없다.

sin과 cos는 최대한 배제하고 transport theorem을 사용한다



b

모든 한 planar에 존재, 이들 사이는 orthogonal이 하지 않으므로 외적 사용.



matrix 사용해서 외적

사용해도 괜찮다

$$\hat{e}_2 = \cos\theta \hat{b}_1 - \sin\theta \hat{b}_3$$

$$\hat{e}_2 \times \hat{b}_3 = (\cos\theta \hat{b}_1 - \sin\theta \hat{b}_3) \times \hat{b}_3$$

$$= -\cos\theta \hat{b}_2$$

하나의 방법일 뿐